

Marking of parts

with trademark and part identification data

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. Author, approver and reviewer are stored in and can be retrieved from the TERE system master data.

Amendments

The following amendments have been made to GS 91001:2020-02:

- section 3 restructured;
- sections 6.6.1 and 6.6.2 added;
- section 7.1 updated;
- section 8 updated;
- section 9 added;
- dates corrected in Figure 9;
- Group Standard editorially revised.

Previous editions

BMWN(S) 11319.0: 1966-09, 1977-09, 1978-03, 1979-02, 1991-07, 1999-09

GS 91001: 1999-09, 2000-12, 2001-12, 2003-06, 2007-04, 2011-12, 2013-11, 2018-09, 2019-08, 2020-02

1 Scope and purpose

All vehicle parts, purchased and/or spare parts and original accessories of BMW Group and associated companies are labeled with a part identification marking. This applies also for flaps, doors and bolted side panels made of sheet metal.

Content of this standard is the marking of parts with the following objectives:

- quick and unambiguous part identification;
- material separation according to substances for recycling and disposal;
- quality improvement by avoiding wrong assembly;
- reduction of warranty costs (restriction in case of error; part tracking);
- marking without disturbing the overall appearance;
- consideration of country-specific and of approval-specific regulations.

Except for

- standard parts (e.g. mechanical fasteners) and standard-similar parts,
- parts not pertinent for homologation, which cannot be marked due to technical reasons (function, quality, surface) and/or due to lack of space.

NOTE Part identification markings such as type-approval marks that shall be applied because of approval certificates are not regulated in this Group Standard. Such markings are subject to the respective legal regulations.

Validity: Changes in this Group Standard compared to the previous issue shall be implemented with the new release at the latest, except for emission-relevant components (section 7).

2 Normative references

This standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

FATEV EG-8/139/2018 ¹⁾	Kennzeichnung typzulassungsrelevanter Bauteile
GS 91002	Marking of parts; Trademark for marking parts
GS 91003	Marking of parts; Marking of material
GS 91004	Marking of parts; Marking with date of manufacture
DIN EN ISO 3166-1	Codes for the representation of names of countries and their subdivisions; Part 1: Country codes
ISO 1073-1	Alphanumeric character sets for optical recognition; Part 1: Character set OCR-A; Shapes and dimensions of the printed image
VDA 260	Components of motor vehicles; Marking of material

1) can be obtained in the releasing specialist department

3 Abbreviations and Terms

3.1 Abbreviations

AI Change index

CCC Product certificate as prerequisite for the import to China (China Compulsory Certificate)

3.2 Terms

Production batch	A production batch starts with the initial part release and ends with the final part. A production batch is the defined quantity of a product (either initial product, intermediate product or end product), which has uniform characteristics and was manufactured in one uninterrupted production cycle (with one material batch). A production batch has the following characteristics: – produced in one work cycle in a specific part of a continuous process; – a specified quantity of material produced using the same starting materials and under the same conditions, for which a uniform nature and quality within predetermined limits is intended and can be regarded as homogeneous.
Serial/part number	The serial or part number is defined in section 6.6.1. Together with the part number and the change index, the part and its origin can be clearly identified. The definition regarding the serial number shall apply only to those scopes which have not already been regulated elsewhere. These scopes shall be agreed separately with the planning departments concerned.

4 General information

Marking shall be carried out such that the function of the parts is not impaired. The durability and readability shall be ensured under normal use of the vehicle over the service life of the part.

The complete marking is to be applied such that it is not directly visible and/or does not disturb the harmony of visual perception in installed position. However, the marking shall be readable, if necessary, with the aid of auxiliary means (e.g. mirror).

Part markings in the engine compartment, which are visible from the top are not permissible, except for those trademarks approved within the scope of the engine compartment styling. Information labels and markings which are legally stipulated or required for reasons of product liability are excluded.

Spare parts with manufacturer's trademark and/or part number of the supplier are always equipped with an additional marking according to this standard. The marking shall be applied with sufficient spatial separation in order to avoid confusion.

In the absence of legal requirements, however, the objective is not to have any manufacturer marking on the part.

Type and material of the part determine the type of application of the marking, e.g. etching, stamping, engraving, embossing, casting, printing.

In case of blanks the marking shall be applied on surfaces which are not machined.

In case the part identification is changed during the series process, the specialist department responsible for release decides, in cooperation with the plants, when the conversion will take place, based on the tool costs.

5 Responsibilities

The necessity and design of the marking is determined by the design department in coordination with the supplier and a respective note is added on the drawing (see section 11).

6 Layout

6.1 General information

A marking consists of:

- trademark (e.g. BMW, MINI, ROLLS-ROYCE);
- parts identification number (supplier-specific/-neutral);
- supplier number and supplier's plant code;
- country of origin and/or country code;
- material code;
- date of manufacture.

If all below mentioned prerequisites are met, the following markings can be omitted (to be evaluated individually):

- part identification number (especially 7-digit part number and change index);
- supplier number and supplier's plant code;
- date of manufacture.

The prerequisites are:

- there is no technical opportunity for a direct marking or a marking is not economically sensible;
- the part shall be clearly identifiable by means of an appropriate IT system (e.g. production database with Data-Matrix code reader);
- the title block of the component shall make a reference to the IT system in which the marking information is stored;
- the packaging shall be marked with the part identification number.

If a part can only be partly marked due to restricted space, marking with the part identification number (supplier neutral part number) and supplier's number/supplier's plant code is to be striven for. In the case of flaps, doors and bolted side panels made of sheet metal a marking with the trademark is sufficient.

6.2 Trademark

Parts are marked with the trademark of the relevant vehicle (e.g. BMW, MINI) into which they are assembled.

In case of original BMW parts in BMW vehicles the BMW symbol shall be preferred.

Common parts used in different brands of the BMW Group are marked with the BMW Group wordmark (= group trademark).

If BMW parts marked with the BMW symbol are used in other brands (adopted parts), then the symbol shall be replaced by the BMW Group script lettering (if economically tenable).

If parts marked with a trademark of a brand other than BMW are installed in a BMW vehicle, the trademark shall be replaced by the BMW wordmark.

Parts which are directly visible shall be marked with the respective trademark of the vehicle for reason of the brand identity.

Engine parts affecting the engine as a unit keep the trademark of the engine manufacturer in case of a potential adoption by other makes.

The trademark nameplate on the cylinder head cover depends on the respective styling and is regulated by the relevant vehicle manufacturer.

For dimension and representation of the trademark refer to GS 91002.

6.3 Part identification number

6.3.1 General information

Lettering shall be carried out in Helvetica standard or according to ISO 1073-1.

Recommended lettering height h is 3 mm to 8 mm.

The approximately required space can be calculated according to the following formula:

$$l = 0,7 \times h \times n$$

Where

l length of line;
 h letter height / lettering height;
 n number of letters.

Spacings between words are counted as letters.

For improved legibility the numbers are grouped one below the other (see section 6.3.2, Figure 1).

In case of limited space, the part number takes precedence over the U.P.G. number.

Information about the part number and the change index (German abbreviation: AI) are submitted to the supplier by the customer (e.g. via order, drawing).

6.3.2 Parts with single part number (as a rule)

Parts that can be identified unambiguously by a single part number obtain one part identification number.

The part identification number is composed of the initial unit parts group (U.P.G. No.), the 7-digit part number (drawing number) and the change index (AI).

In order to ensure the fulfillment of country-specific regulations it is necessary to display the initial U.P.G. and the change index spatially separated from the part number. Generally, the initial U.P.G. is written in an individual line above the part number and the change index is written in an individual line below the part number (Figure 1 and section 11, Figure 7, shape A). If this is not possible due to restricted space, a minimum distance of $3 \times h$ (h = lettering height) shall be observed both between the initial U.P.G. and the part number and between the part number and change index (example: section 11, Figure 7, Shape B)

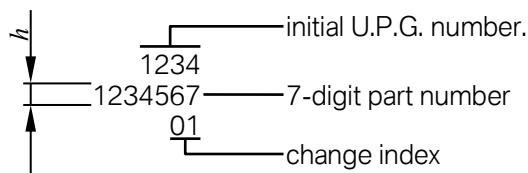


Figure 1: Part identification number

In order to reduce costs of tool changes for components that require identification of the tool with the change index (e.g. rubber elements), it is permitted to apply, from the outset, several change indices separated by vertical lines (Figure 2). The number of the applied change indices shall be agreed between the customer (BMW) and the supplier.

Invalid change indices shall be identified by an "X". The first non-marked change index is the currently valid change index AI (Figure 2).

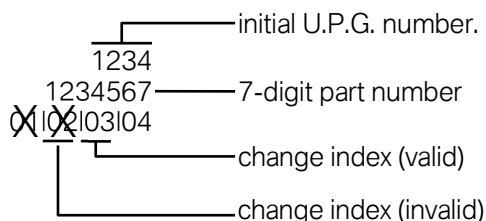


Figure 2: Part identification number with several change indices

6.3.3 Parts with two part numbers

Parts that are interchangeable from a functional view or a manufacturing view and to which one part number per supplier has been assigned, obtain a supplier-specific part number “LS” and a supplier-neutral part number “LN”.

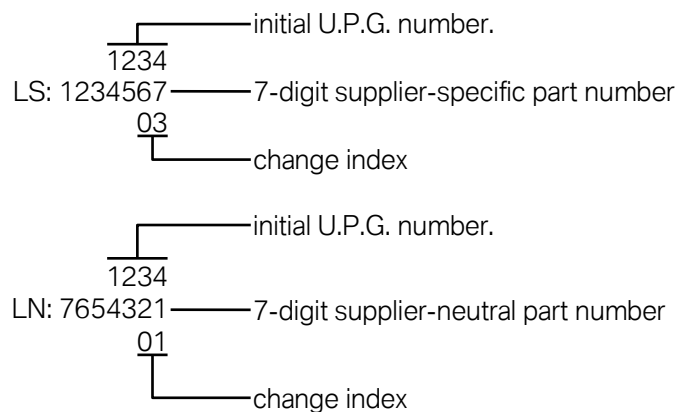
The supplier-specific part number “LS” is assigned to a certain supplier and with that number the customer service can identify the supplier in a warranty case. If possible, it shall be highlighted (larger lettering and/or bold).

The supplier-neutral part number “LN” along with the supplier number / supplier’s plant code are used for handling the logistical processes.

If a part can be marked only partly for reasons of restricted space, at least the supplier-neutral part number and a manufacturer characteristic shall be applied on the part.

NOTE When marking batteries, the change index is not taken into account.

Figure 3: Part identification number with supplier-specific and supplier-neutral part number



6.4 Manufacturer's name

Lettering shall be carried out in Helvetica standard or according to ISO 1073-1.

The manufacturer's name shall be applied to the part in capital letters and shall not contain any abbreviations.

Recommended lettering height h is 3 mm to 8 mm.

6.5 Supplier number / supplier plant code

The 6-digit number of the supplier is a BMW-specific identification of the supplier.

Marking with the supplier number is required in order to monitor the quality and to be able to identify the manufacturer/place of manufacture if warranty or product liability cases should occur.

The 2-digit plant code encodes the manufacturing site. Separated by a space, the plant code is added to the supplier's number.

The marking is to be applied below or adjacent to the part identification number. Lettering and lettering height correspond to the part identification number.

6.6 Serial number, part number and production batch

6.6.1 Serial number or part number

The serial number or part number contains the following in coded form, e.g.:

- date of production of a part;
- shift counter;
- assembly location;
- information on manufacturer;
- information on customer;
- day production counter.

If an index change is required with the same serial number the day production counter may not be restarted.

The serial number shall be unique.

The position of the marking of serial number or part number shall be indicated in the 3D model or the drawing.

6.6.2 Production batch

The position for the marking of the production batch shall be indicated in the 3D model or the drawing.

6.7 Country of origin (CO)

Due to existing laws and/or customs regulations, vehicle parts which are exported as single parts, aggregates, spare parts or accessories shall be marked with the name of the country of origin.

Parts are marked with the official English country designation according to DIN EN ISO 3166-1 or according to the online browsing platform of the International Organization for Standardization (<https://www.iso.org/obp/ui/#search> → Country codes) in capital letters, e.g. GERMANY. If space is limited the Alpha-2 code is sufficient. Lettering and lettering height correspond to the part identification number.

6.8 Material code

Vehicle parts are provided with a material code for identification of the material.

This is the prerequisite for:

- the material separation according to substances for recycling and disposal;
- the processing of reusable materials;
- the use of parts for repair work.

The basis for marking is VDA 260. BMW-internal specifications are governed in GS 91003.

Marking with the material code shall be specified on the drawing.

6.9 Date of manufacture

Vehicle parts, which shall bear the manufacturing date to proof the date of manufacture, shall be marked in accordance with GS 91004.

The type and shape of the manufacturing date marking is defined by the design department and specified on the drawing.

7 Emission-relevant components

7.1 General information

If emission relevant parts are to be marked due to type approval requirements or legal specifications with the manufacturer's name in capital letters, the marking shall be applied according to the following example.

_____ initial U.P.G. number.
1234
1 234 567 _____ 7-digit part number
01 _____ change index
TENNECO
_____ Manufacturer's name

Figure 4: Marking with manufacturer name (exemplary)

7.2 Marking of catalytic converters

Catalytic converters shall be marked as follows:

_____ part number
1 234 567
01 _____ change index
TENNECO _____ manufacturer of catalytic converter
SEALING: TENNECO _____ manufacturer canning and catalytic converter
CARRIER: NGK _____ manufacturer of carrier
COATING: UMICORE
_____ manufacturer of coating

Figure 5: Marking of catalytic converters (exemplary)

Alternatively the following marking is possible:

_____ part number
1 234 567
01 _____ change index
TENNECO _____ manufacturer of catalytic converter
FZ: TENNECO _____ manufacturer canning and catalytic converter
ZT: NGK _____ manufacturer of carrier
TC: UMICORE
_____ manufacturer of coating

Figure 6: Alternative marking of catalytic converters (exemplary)

8 Type-approval identification for China

For the China homologation a type-approval identification of the pertinent components as well as an identical entry of the identification shall be made in the approval documents for vehicle projects offered in China.

Marking of emission-pertinent and C CC-pertinent components are described in sections 7 and 10. In addition to the regulations for CCC and emissions, further parts are affected by legal and type approval-relevant specifications, these shall be obtained from FATEV EG-8/139/2018. Pertinent components shall be inquired for at the Type Approval department in case of uncertainties.

The type-approval identification shall be positioned in a separate line. If this is not possible, a minimum space of $3 \times h$ (h = lettering height, see section 6.3.2) before and after lettering shall be complied with (example: section 11, Figure 7, shape B). If the space is restricted the use of a frame around the identification is permitted.

If the part numbers is used as type-approval identification, it is mandatory to write the AI in its individual line underneath the part number.

The type-approval identification shall be created and administrated by the releasing department. Should a type-approval relevant change of the component cause a new homologation or re-homologation, the change of the type-approval identification as well as the adaptation of the approval documents shall be realized analogous. An adaptation of this identification without type-approval relevant change at the component is not permissible.

The type-approval identification serves to identify the components for the Chinese authorities. Hence, the identification shall be 100 % identical to the specification in the approval document (upper case, lower case, special characters, blanks and digits).

NOTE A standardization of the type-approval identification per subject (nomenclature) is recommended and should be created by means of characters, capital letters and without special characters.

EXAMPLE Passenger Airbag PAB-G30-01

Table 1: Possible nomenclature for a type-approval identification

Airbag component	Abbr.	Technical initial derivative	Production	Abbr.	Homologation index	DAB variant	Abbr.
Passenger airbag	PAB	G30 (example)	China local	CN (only in case of China localization)	01	Base (AG)	AG-B
Driver Airbag	DAB	F90 (example)			02	Sport (AG)	AG-S
Knee airbag	KAB				ff.	M-Sport	M
Curtain airbag	CAB						
Side airbag (front)	SAF						
Side airbag (rear)	SAR						
Pelvis pusher	PEP						

The use of the BMW development name only (e.g. G08) is not permitted as type approval identification.

9 Part marking for remanufactured spare parts

Remanufactured spare parts shall be marked with the following lettering directly on the part (assemblies are regarded as one spare part):

RP REMAN

The lettering RP REMAN means “returnable part remanufactured”. The marking shall be applied in capital letters, complete and without further abbreviations. The requirements set forth in section 4 shall be fulfilled.

The original part identification number shall be removed.

The remanufactured part shall be marked with a new part identification number as specified in section 6.3. The marking shall comply with the requirements in section 4 and with the original marking (marking of series production part and marking RP REMAN). The position of the marking shall correspond to the position of the original marking.

Remanufactured parts shall be marked accordingly on the packaging, the invoice and the repair request.

10 Component marking for type approvals and product certificates

For the homologation of vehicles in the different markets of the world, for certain parts and components, type-approvals (equal to product certificates, e.g. CCC certificates, ECE type approval, AIS certificates) shall be provided to BMW. The supplier is responsible for taking the necessary actions to obtain the type approvals. Type approval shall be ensured by supplier.

Depending on the type approval, specific markings shall be applied. The supplier shall obtain the required information concerning the specific marking and shall ensure the application of the marking.

Since the marking serves as identification for the respective authorities it shall be 100 % identical with the respective type approval and type approval documents. This especially applies to (upper case, lower case, special characters, blanks and digits. Furthermore, proper readability on the component shall be ensured.

The note regarding marking relevancy is made on the drawing by the releasing department. Components to be tested in the course of a total vehicle approval or as single part shall comply with the approved (tested) series production status.

Changes to the part marking, referring to the type approval, require an update of the type approval documents. Hence, when planning such a change the “Type Approval, Development Abroad, Conformity of Production” (passenger car and motorcycle) specialist department shall be incorporated.

NOTE A standardization of the model identification per subject (nomenclature) is recommended and should be created by means of characters, capital letters and without blanks.

11 Design

Type, design and arrangement of the marking are determined by the design department in coordination with the supplier.

Other shapes and arrangements or additional data shall be represented directly in the drawing.

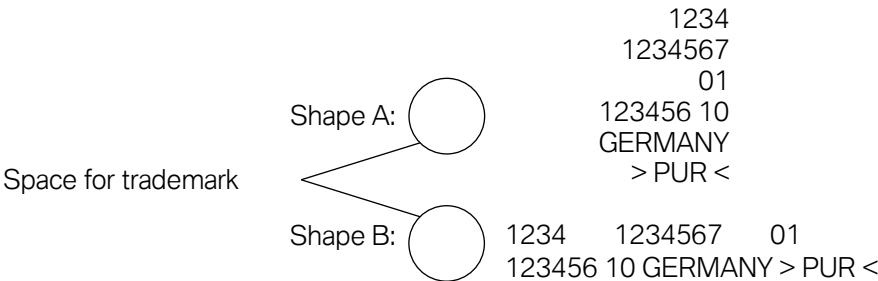


Figure 7: Version (as a rule)

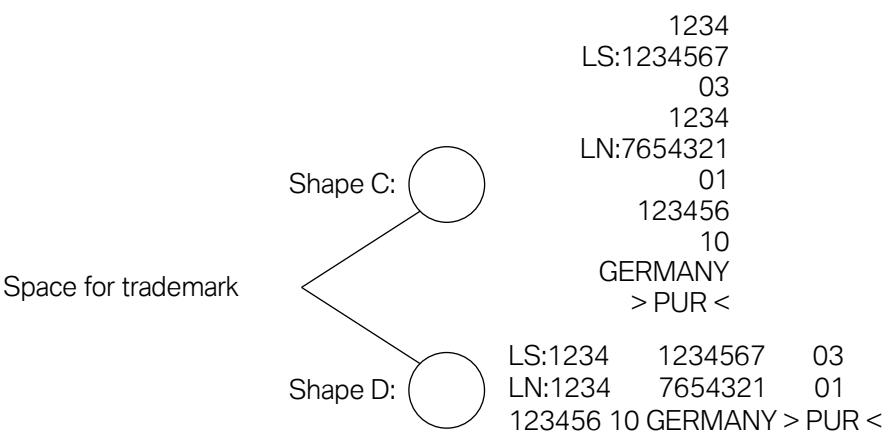


Figure 8: Version (supplier-specific parts)

The following example shows the maximum part identification marking in vertical and horizontal direction (except for catalytic converters, refer to section 7.2).

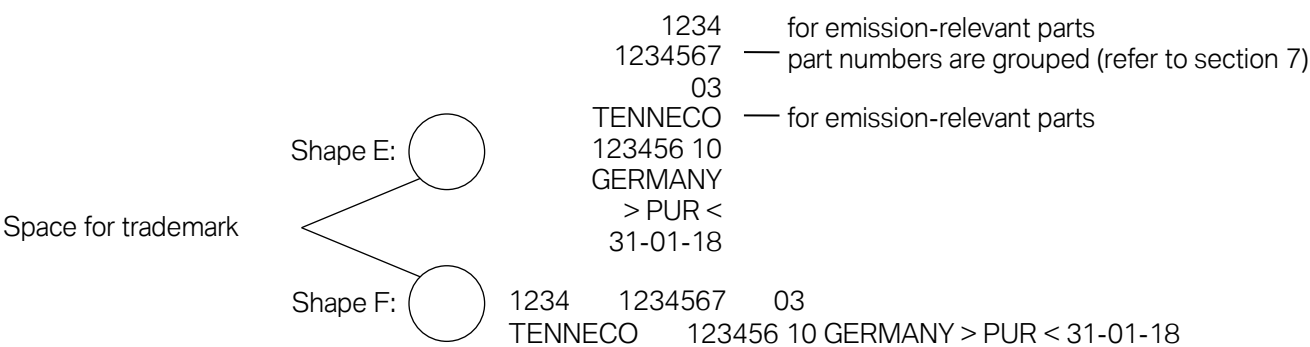


Figure 9: Maximum part identification marking

The following example shows the minimum part identification marking in vertical and horizontal direction (except for catalytic converters, refer to section 7.2).

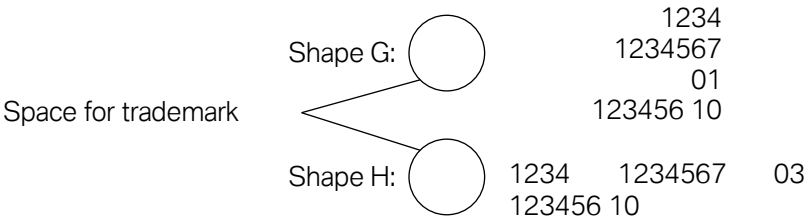


Figure 10: Minimum part identification marking

12 Drawing entry

The drawing entry shall be done in the appropriate view with a reference line to a marked surface.

If several change indices are applied from the outset (see Figure 2), the number of the indices shall be indicated as described in example 3.

The trademark, the identification data and the manufacturing date are separately marked.

EXAMPLE 1

- trademark BMW symbol
- identification data shape A, lettering height 4
- supplier number
- country of origin (CO)
- material PUR
- date of manufacture

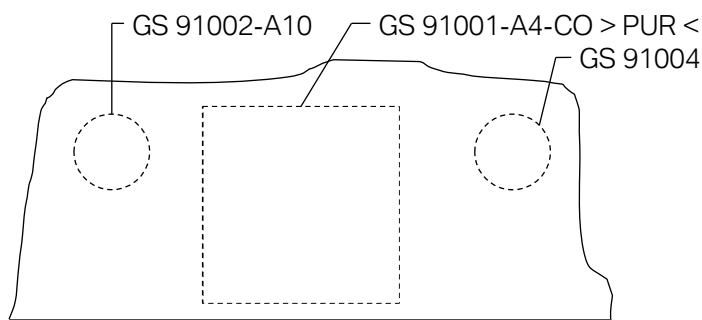


Figure 11: Marking with trademark, identification data and date of manufacture (drawing entry)

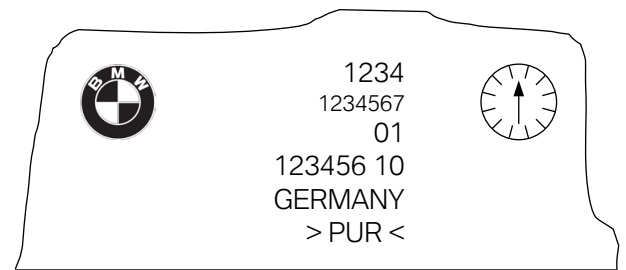


Figure 12: Marking with trademark, identification data and date of manufacture (exemplary)

EXAMPLE 2

- trademark MINI wordmark
- identification data shape B, lettering height 5
- country of origin
- material EPDM

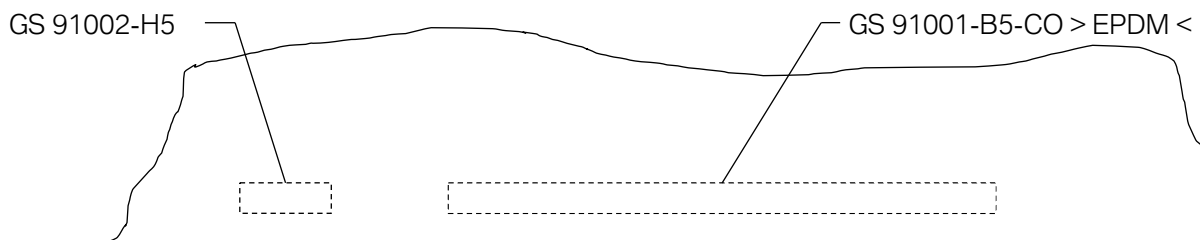


Figure 13: Marking with trademark and identification data (drawing entry)

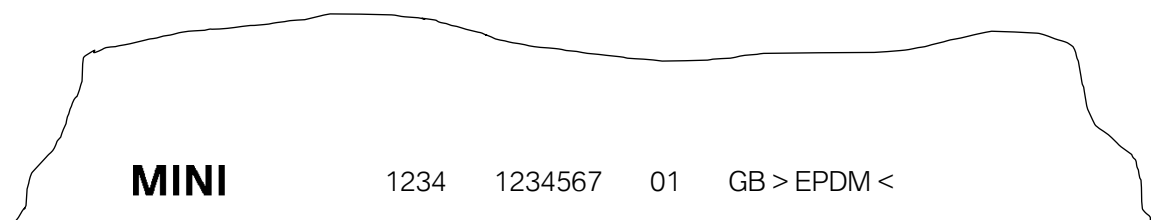


Figure 14: Marking with trademark and identification data (exemplary)

EXAMPLE 3

- trademark BMW symbol
- identification data shape A, lettering height 4
- number of AI: 5
- supplier number
- country of origin (CO)
- material PUR
- date of manufacture

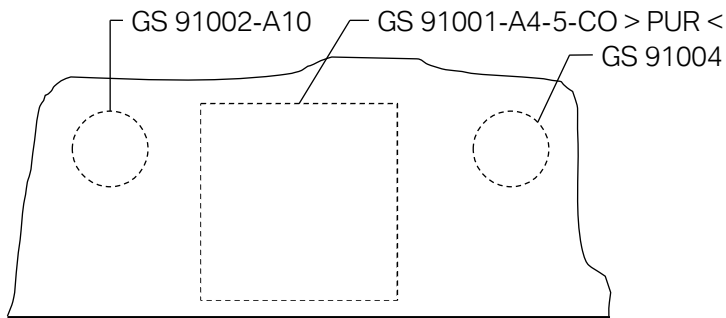


Figure 15: Marking with trademark, identification data and date of manufacture (drawing entry)

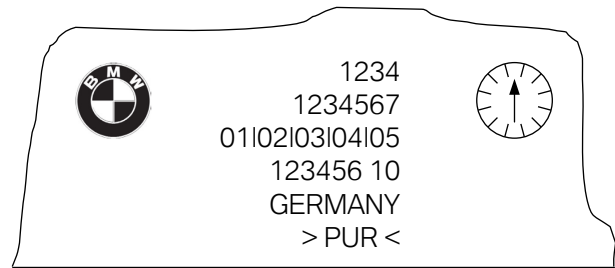


Figure 16: Marking with trademark, identification data and date of manufacture (exemplary)

13 Approval

For the approval, the design of the initial sampling is authoritative.

Each change shall be introduced by a new sampling and requires a new written release.